

THE KEY TO SUCCESSFUL SEASONS IN PRO CYCLISTS

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In past KIW Research Notes and many times in general, we learn about cycling science performance from ‘*controlled*’ *training studies*. What they are? They consist in dividing cyclists into different groups which perform different standard controlled training protocols. Sometimes, these studies are criticized because, in order to isolate one training variable, they don’t permit to study training in ‘real-world’ scenario.

Retrospective training analysis represent an interesting alternative way to learn. They ‘just’ look back at training characteristics of certain athletes and see what was more and less successful. In this study design, you can see a training concept not isolated, but already in the mix with all the ‘real-world’ variables. Please, note that controlled studies are still a very good method to understand principles and apply them adapting to your own situation: they should just not be used as copy and paste. Retrospective analysis of training characteristics are not better or worse than controlled training studies, they are an interesting alternative way to learn.

Which training and racing characteristics (volume, frequency, load, intensity, time spent in zones, ...) are associated with more successful seasons in professional road female cyclists?

A hot off the press retrospective training analysis published by Annemiek Roete and Colleagues (University of Groningen, Netherland) in 2025 on *International Journal of Sports Physiology and Performance* tried to answer this question. (1)

WHAT DID THEY DO?

- Training and racing characteristics (volume, frequency, load, intensity, time spent in zones, ...) of 43 seasons of 14 World Tour female cyclists were retrospectively analysed. Time spent in zones (hours) was evaluated through the following 5 zones model: zone 1: $\leq 55\%$ FTP, zone 2: 56-75% FTP, zone 3: 76%-90% FTP, zone 4: 91%-105% FTP, and zone 5: $\geq 106\%$ FTP.
- The time frame analysed was 2013–2019. During this period the participants’ team finished every year within the top 10 of the UCI team ranking.
- Based on the ProCyclingStats points per race, cycling seasons were classified as:
 - **Highly Successful seasons (HS):** if ≥ 5 points per race were scored.
 - **Less Successful seasons (LS):** if < 5 points per race were scored.

WHAT DID THEY FIND?

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- **Length of the season, total number of training sessions and race days were similar between Highly and Less Successful Seasons:**

Length of the season: 328 ± 17 vs 335 ± 10 days;

- Total training days: 219 ± 18 vs 204 ± 26 ;
- Race days: 45 ± 7 vs 46 ± 11 ;

- **Total work done per training session was higher in Highly vs Less Successful Seasons:**

Work done per training session: 1354 ± 185 vs 1164 ± 114 kilojoules;

The higher work done was the result of a higher training volume but not intensity performed during HS vs LS seasons (see below).

- **Total season exercise duration was higher in Highly vs Less successful seasons:**

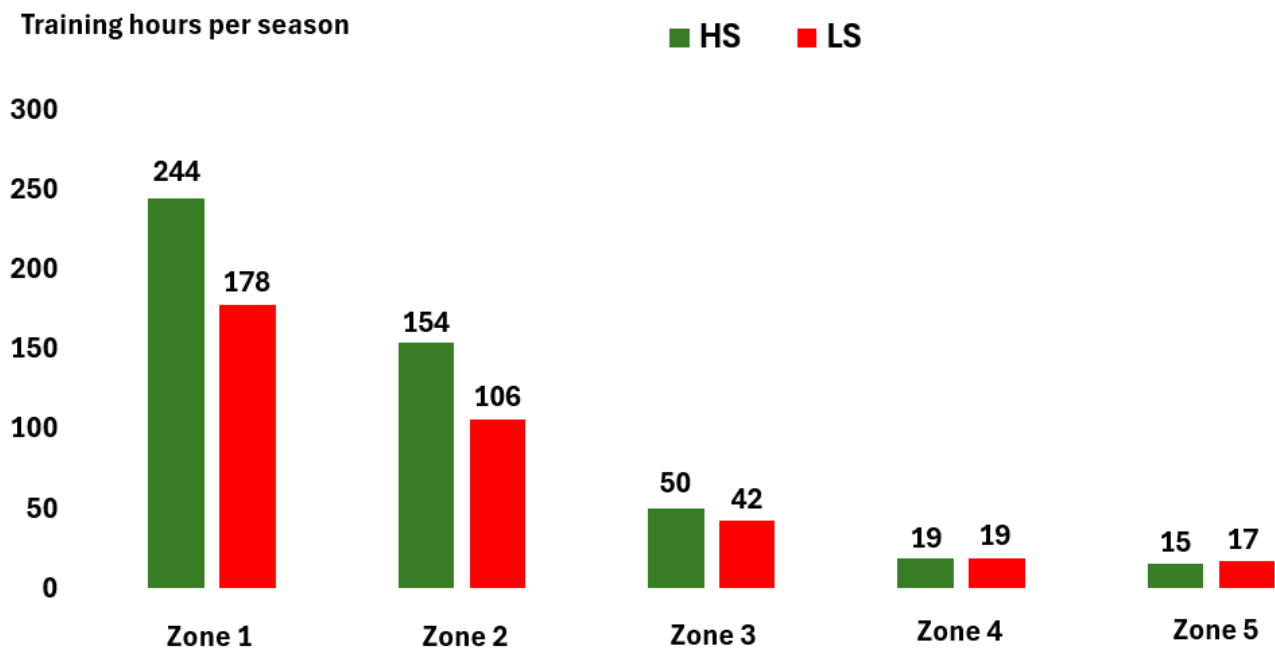
Training + racing total duration: 563 ± 62 vs 443 ± 81 hours;

This was the result of more volume done in training and not in racing in HS vs LS:

- Training duration: 560 vs 469 hours;
- Racing duration: 144 vs 133 hours;

- **For what concern absolute time spent in zones (hours) during training sessions, time accumulated at low intensity (zone 1 and zone 2) was higher in HS vs LS, time spent at medium intensity (zone 3) tend to be higher in HS vs LS, while time spent at high intensity (zone 4 and 5) was the same between the two groups:**

- Zone 1: 244 vs 178 hours;
- Zone 2: 154 vs 106 hours;
- Zone 3: 50 vs 42 hours;
- Zone 4: 19 vs 19 hours;
- Zone 5: 15 vs 17 hours;



PRACTICAL TAKEAWAYS

A higher volume of low intensity training is the only exercise-related variable which differs between highly and less successful seasons in professional World Tour female cyclists.

GABRIELE'S COMMENT

As the authors of the study noted:

“These findings highlight the importance of high-volume low-intensity training in female professional cyclists”

- Why a higher volume of low intensity training is associated with more successful season?

Here my interpretation (based on scientific data):

- A high volume of low intensity training seems to be important to optimize durability (fatigue resistance/resilience during prolonged exercise).
- Even if, to stimulate training adaptations, high intensity training is more powerful than low intensity training when the total energy expenditure is matched, the amount of high intensity training doable within a microcycle is much more limited both from a physical (cardiac autonomic recovery, HRV) and psychological point of view. So, after already doing a certain amount of weekly high intensity training, the only way to further improve endurance performance level in a long term sustainable way is to progressively increase the volume at low intensity.

- High volume of low intensity training: How high it should be?

Of course, these data should not be read as 'you should try to immediately jump on the highest volume you can do passing from 10 to 25 hours per week'. In order to give a progressive overload without distressing the body, I would suggest to increase training volume by no more than ~10% per season. In fact as the authors of this study noted, '*achieving high volumes requires multiple years of dedicated training, which is reflected in the higher overall age of the athletes recorded in highly vs less successful seasons*': 25 ± 3 vs 23 ± 5 yrs.

Thanks for Reading! Share with your Team!

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REFERENCE

1. Roete AJ, van Erp T, Elferink-Gemser MT, Kempe M, Lamberts RP. Internal and External Load Characteristics in Professional Female Cyclists During Highly Successful and Less Successful Seasons. Int J Sports Physiol Perform. 2025 Aug 5;20(9):1224-1231.



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